$$\begin{tikzcd}
E & A \arrow[l] \arrow[r, "LongElement"] \arrow[d, "h"'] & B \\
& D & C \arrow[l, "k"] \arrow[u, "g"']
\end{tikzcd}$$

$$\begin{tikzcd}
E \arrow[dr] & A \arrow[l] \arrow[r] \arrow[d] & B \\
& D & C \arrow[l] \arrow[u]
\end{tikzcd}$$

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[dr, "f;g"'] & B \arrow[d, "g"] \arrow[dr, "g;h"] & \\
& C \arrow[r, "h"'] & D
\end{tikzcd}$$

$$\begin{array}{ll}
\text{corner} & \begin{array}{cccc} c_1 & c_2 & \dots & c_n \end{array} \\
\begin{array}{l} r_1 \\ r_2 \\ r_3 \\ r_4 \end{array} & \begin{pmatrix} a_{11} & 0 & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}
\end{array}$$

$$\begin{array}{rrrrcr}
3x_1 & -2x_2 & +x_3 & -x_4 & = & 7 \\
-x_1 & & -5x_3 & +2x_4 & = & 2 \\
& x_2 & +2x_3 & & = & 0 \\
2x_1 & +3x_2 & & -5x_4 & = & -1
\end{array}$$

$$\left(\begin{array}{cccc|c}
3 & -2 & 1 & -1 & 7 \\
-1 & 0 & -5 & 2 & 2 \\
0 & 1 & 2 & 0 & 0 \\
-2 & 3 & 0 & -5 & -1
\end{array}\right)$$

$$\left[\begin{array}{cccc|r}
3 & -2 & 1 & -1 & 7 \\
-1 & 0 & -5 & 2 & 2 \\
0 & 1 & 2 & 0 & 0 \\
-2 & 3 & 0 & -5 & -1
\end{array}\right]$$

$$\sigma(X) = \frac{p\left(x \mid H_1\right)}{p\left(x \mid H_0\right)} \underset{H_0}{\overset{H_1}{\gtrless}} \theta$$

$$\left(\text{Left Label} \left[\begin{array}{cccc} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{array} \right. \begin{array}{l} \left.\begin{array}{l} \\ \\ \end{array}\right\} \text{Right Label} \\ \\ \\ \end{array} \right)$$

$$\cancel{2x + 4y}$$

$$\bcancel{2x + 4y}$$

$$\xcancel{2x + 4y}$$

$$f(x) = \frac{(x^2 + 1)\cancel{(x - 1)}}{\cancelto{1}{(x - 1)(x + 1)}}$$

$$\begin{pmatrix} \begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{matrix} & 0 & 0 \\ 0 & \begin{matrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{matrix} & 0 \\ 0 & 0 & \begin{matrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{matrix} \end{pmatrix}$$

$$\begin{pmatrix} x_R \\ y_R \end{pmatrix} = \underbrace{r}_{\text{Scaling}} \cdot \underbrace{\begin{pmatrix} \sin\gamma & -\cos\gamma \\ \cos\gamma & \sin\gamma \end{pmatrix}}_{\text{Rotation}} \begin{pmatrix} x_K \\ y_K \end{pmatrix} + \underbrace{\begin{pmatrix} t_x \\ t_y \end{pmatrix}}_{\text{Translation}}$$

$$|x| = \begin{cases} x, & \text{for } x \geq 0 \quad \text{(1a)} \\ -x, & \text{for } x < 0 \quad \text{(1b)} \end{cases}$$

$$F(x) = \int f(x)\,\mathrm{d}x + C \quad \ldots\ldots\ldots\ldots \quad (2)$$

$$F(x) = \int_1^\infty f(x)\,\mathrm{d}x = 1 \quad \ldots\ldots\ldots\ldots \quad (3)$$